

Article

Investigating Awareness of Pesticide Exposure as a Risk Factor for Parkinson's Disease and Uptake of Exposure-Mitigating Behaviour in Farming Communities in Ireland

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Abstract: Parkinson's disease (PD) is an age-related neurological disorder with increasing incidence and modifiable risk factors. People exposed to pesticides have up to a 2-fold higher risk of developing PD. Use of personal protective equipment (PPE) when using pesticides can lower an individual's exposure. We examined awareness of the relationship between pesticides and PD risk in individuals working/living on farms in Ireland. We also investigated the practice of behaviours aimed at mitigating exposure, such as using PPE. An online survey was completed by a sample of the farming community ($n = 707$) attending agricultural fairs, and included demographics, lifetime/current residence/work on farms, pesticide contact, PPE use, PD diagnosis, and awareness of pesticide–PD association. Among participants, 51% worked/lived on farms and 62% reported contact with pesticides. Only 69% of those with pesticide contact reported using PPE, with gloves (57%) and masks (50%) most commonly used. Only 22% were aware of an association between PD and pesticides, and awareness did not increase PPE use. Among people with PD, only 40% had knowledge of the risk. We found that in a highly agricultural economy, occupational exposure to pesticides is common, but mitigation behaviours are not optimal. Educational campaigns to improve awareness of health risks from pesticides and to encourage PPE use could lower the personal and healthcare burden of PD and other health outcomes.

Keywords: Parkinson's disease; pesticides; PPE; environmental risk



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1. Introduction

Parkinson's disease (PD) is a neurodegenerative disorder of increasing incidence globally due to changing demographics and a steady rise in the ageing population [1]. PD is characterised by degeneration of multiple neuronal pathways in the brain, particularly dopaminergic (DA) pathways, which control movement; degeneration of these leads to the cardinal motor symptoms of PD. Understanding and mitigating any modifiable causes of PD are important to reduce the incidence and burden of this debilitating disease. Several international studies have found that communities that are exposed to pesticides have higher incidences of PD [1–7]. Organic pesticides such as rotenone are commonly used to produce laboratory models of PD, as they induce pathological pathways in cells that mimic

those that are known to occur in human PD, such as toxic misfolding of alpha-synuclein protein, which is a post mortem pathological hallmark of this disease [8]. These studies provide direct mechanistic evidence for a link between environmental pesticides and PD pathology. In addition, several pesticides have been shown to be directly toxic to DA neurons derived from human-induced pluripotent stem cells (iPSCs) [9], as well as DA neurons in cell culture and animal models of PD [8,10].

Individuals who use pesticides can mitigate their exposure and their potential risk of developing PD through the use of personal protective equipment (PPE) such as masks, coveralls/aprons, gloves, shoes/boots, glasses/goggles, and hats [11]. The use of gloves in particular has been shown to modify or negate the association of pesticides, specifically paraquat, with the risk of developing PD [12]. It has been reported that with increased age, there is less awareness about the safe use of pesticides, and that awareness and use of safety behaviours are also associated with participants' overall education levels [12–14]. Furthermore, farmers who are more aware of the risks associated with pesticide use are more likely to use PPE [14]. Studies have found that the use of PPE by people working with pesticides was low regardless of type or class of pesticide, but it has been highlighted that the use of PPE and overall safety behaviours can be improved with training [12,13]. Targeted educational interventions have been shown to improve safety attitudes and behaviours involved in pesticide use [12].

Ireland has a diverse and rich farming culture, representing all of the major farming classifications. The total number of farms in Ireland is circa 133,000, with the agricultural workforce at almost 300,000 individuals [15]. Farming in Ireland includes all of the main farming types, such as crop, tillage, beef, pigs, and sheep, and only 3.4% of farm holdings were classed as organic in 2020 [16]. The agriculture and food sector represents 9.5% of total exports, with a value of EUR 18.3 billion in 2023 [17]. Therefore, to enhance or protect this asset, pesticides are widely used for crop protection and yield enhancement. The aim of our study was to document the awareness of, and practices related to, pesticide use among a large and high-risk occupational group in Ireland, and to measure the knowledge of this population about the potential link between pesticide exposure and risk of PD.

2. Materials and Methods

A cross-sectional, remote, anonymous, survey-based study was conducted in farming communities across Ireland, primarily by the recruitment of participants in 2022 at three national and local agricultural shows and competitions, as well as via online recruitment through social media. Inclusion criteria were individuals aged 18 years and over with fluent English. Participants included males and females who actively engaged in agricultural activities and/or lived in rural communities (total $n = 707$). Ethical approval for this project was granted through the University College Cork (UCC) Social Research Ethics Committee (SREC), Log number: 2021-116. Survey design, distribution, and data collection were conducted anonymously using the REDCap electronic data capture tool, hosted at UCC. Survey Questions are included as Supplementary Material. REDCap (Research Electronic Data Capture) is a secure, web-based software platform designed to support data capture for research studies, providing (1) an intuitive interface for validated data capture; (2) audit trails for tracking data manipulation and export procedures; (3) automated export procedures for seamless data downloads to common statistical packages; and (4) procedures for data integration and interoperability with external sources [18].

Data were collected using structured questionnaires; the duration of completion was between 10 and 15 min. The questionnaires included sections on demographics including gender, age, tobacco-smoking behaviour, PD diagnosis and family history, farming history, including whether the respondent currently lived or worked on a farm, or if they grew up

on a farm, as well as the duration of time living on a farm, the type and size of farm, and the types of pesticides used. Participants that confirmed that they had used pesticides were then asked about the class (insecticide, herbicide, fungicide, or do not know), name (open text), frequency (days per year, open text) and duration of use (1–5, 6–10, 11–15, >15 years) and about the safety practices that they used to mitigate their pesticide exposure, including questions about using several specific items of PPE (masks, gloves, apron, boots, or none). Participants were also asked about their awareness of health risks associated with pesticide exposure, including risk of PD (“Are you aware of the link between Parkinson’s disease and pesticide use?”). To preserve anonymity, the location at which surveys were completed was not tracked. Participants were not provided with incentives.

Data were converted to a Statistical Analysis System (SAS) © format for descriptive analysis. Only data from respondents who had consented to the use of their responses and who were aged over 18 years were included in this analysis.

3. Results

3.1. Participant Demographics

This study included 707 participants from diverse rural backgrounds throughout Ireland. More participants were female (54.9%). Participant ages ranged from 18 to 89 years, with most respondents aged 50–69 years (56%). Most people reported that they had never smoked (65.4%). In terms of farming background, 69.4% of people did not grow up on a farm. While 51.3% of participants were currently working or living on a farm, 62% reported that they came in contact with pesticides at work or home. A diagnosis of PD was reported by 68 (9.6%) respondents (Table 1).

Table 1. Demographic profile of respondents to survey on pesticides and Parkinson’s disease ($n = 707$).

All Respondents <i>N</i> = 707	In Contact with Pesticides	
	Yes (<i>N</i> = 438)	No (<i>N</i> = 244)
Gender	<i>N</i> (%)	<i>N</i> (%)
Male	308 (43.6)	71 (29.1)
Female	388 (54.9)	171 (70)
Non-binary	1 (<1%)	-
Transgender	1 (<1%)	1 (<1%)
Not specified	9 (<2%)	-
Age		
≤20	20 (<3%)	9 (3.7)
21–49	195 (27.6)	79 (32.4)
50–69	396 (56.0)	131 (53.4)
70–89	87 (12.3)	25 (10.3)
Not specified	9 (<2%)	-
Smoking status		
Never smoked	462 (65.4)	151 (61.9)
Former	186 (26.3)	73 (29.9)
Current smoker, <daily	21 (3.0)	11 (4.5)
Current smoker, daily	28 (4.0)	9 (3.7)
Grew up on a farm		
Yes	221 (31.3)	124 (50.8)
No	459 (64.9)	120 (49.2)
Unanswered	27 (3.8)	-
Currently living/working on farm		
Yes	363 (51.3)	86 (35.3)
No	317 (44.8)	157 (64.3)
Unanswered	27 (3.8)	1 (<1)

Table 1. *Cont.*

All Respondents	In Contact with Pesticides		
N = 707	Yes (N = 438)		No (N = 244)
Contact/using pesticides at home/work			
Yes	438 (62.0)	438 (100)	
No	244 (34.5)		244 (100)
Unanswered	25 (3.5)		
Diagnosed with Parkinson's Disease			
Yes	68 (9.6)	43 (9.8)	25 (10.3)
No	599 (84.7)	387 (88.4)	212 (86.9)
Unanswered	40 (5.7)	8 (1.8)	7 (2.9)

3.2. Pesticide Use and Safety Practices

Among the respondents to our survey, 438 (62.0%) reported that they “come in contact with or use” pesticides at work or home. Among the 438 who reported contact with pesticides, herbicides was the most frequent type used (by 83%), followed by insecticides (30%) and fungicides (22%). Most participants who were in contact with pesticides (69.4%) reported using at least one piece of PPE (Table 2). Gloves and masks were the most commonly used items of PPE (56.6% and 50.2%, respectively) (Table 2).

Table 2. Characteristics and use of PPE among those survey respondents who reported having contact * with pesticides (n = 438).

Gender	N (%)	Currently Working/Living on Farm	N (%)
Male	210 (48.0)	Yes	277 (63.2)
Female	227 (51.8)	No	160 (36.5)
Non-binary	1 (<1)	Unanswered	1 (<1)
Age (years)		Use of PPE	
≤20	10 (<3)	Yes (at least one piece)	304 (69.4)
21–49	112 (25.6)	Mask—Yes	220 (50.2)
50–69	256 (58.5)	Gloves—Yes	248 (56.6)
70–89	59 (13.5)	Overalls—Yes	74 (16.9)
Not specified	1 (<1)	Aprons—Yes	20 (4.6)
		Boots—Yes	153 (34.9)
		No (do not use PPE)	126 (28.8)
Diagnosed with PD		Aware of link between pesticides and PD	
Yes	43 (9.8)	Yes	101 (23.1)
No	387 (88.4)	No	330 (75.3)
		Unanswered	7 (2)

* Respondents were asked “Do you come in contact with or use pesticides in your work or at home?”.

3.3. Awareness of Link Between Pesticide Use and Risk of Parkinson's Disease

When participants (n = 671 answered this question) were asked about the health risks associated with pesticide exposure, most respondents (n = 522, 78%) were not aware of the link between pesticide exposure and risk of PD. The level of awareness by gender, age, or smoking status are shown in Table 3. People who reported being in contact with pesticides were no more aware of the link than those who did not work with pesticides (77 vs. 80%). Similarly, use of PPE did not significantly differ according to awareness. However, among

people with PD, 40% (27/67) reported awareness of the association with pesticide exposure, compared to 22% of overall participants (Fisher's exact; $p = 0.0002$) (Table 3).

Table 3. Knowledge of association between pesticides and Parkinson's disease (* $n = 671$).

Aware of Risk ($n = 149$, 22%)		Unaware of Risk ($n = 522$, 78%)
Gender	N (%)	N (%)
Male	64 (48.0)	230 (44.1)
Female	83 (51.8)	292 (55.9)
Transgender	1 (<1)	-
Not reported	1 (<1)	-
Age (years)		
≤20	1 (<1)	18 (3.5)
21–49	44 (29.5)	145 (27.8)
50–69	90 (60.4)	290 (55.6)
70–89	14 (9.4)	68 (13.0)
Not specified	-	1 (<1)
Smoking status		
Never smoked	103 (69.1)	340 (65.1)
Former	41 (27.5)	140 (26.8)
Current smoker, <daily	2 (<2)	18 (3.5)
Current smoker, daily	3 (2.0)	24 (4.6)
Diagnosed with PD		
Yes	27 (18.1)	40 (7.7)
No	121 (81.2)	478 (91.6)
Unanswered	1 (<1)	4 (<1)
Currently working/living on farm		
Yes	87 (58.4)	272 (52.1)
No	61 (40.9)	250 (47.9)
Unanswered	1 (<1)	-
In contact with pesticides at work/home		
Yes	101 (67.8)	330 (63.2)
No	48 (32.2)	192 (36.8)
Unanswered	-	
Use of PPE equipment (** $n=431$)		
Yes (at least one piece)	72 (71.3)	229 (69.4)
Mask—Yes	54 (53.4)	164 (50)
Gloves—Yes	63 (62.4)	183 (55.5)
Overalls—Yes	16 (15.8)	57 (17.3)
Aprons—Yes	5 (4.9)	15 (4.5)
Boots—Yes	32 (31.7)	121 (36.7)
No (do not use PPE)	28 (27.7)	98 (29.7)

* The remainder of respondents did not answer the question on awareness of a link between pesticides and PD ($n = 36$). ** Only respondents who reported 'Yes' to contact with pesticides were asked about PPE.

4. Discussion

A link between pesticide use and the risk of developing PD is well established in epidemiological studies conducted in several other countries which have high levels of agricultural activity [1–7]. It has previously been reported that individuals with PD are

significantly more likely to report direct exposure to pesticides compared with their relatives who do not have a PD diagnosis [19].

Our study is the first to measure awareness of this association within farming communities in Ireland. We report that the overwhelming majority of our survey participants were unaware of the hazardous link between pesticides and PD, regardless of whether they reported regular use of pesticides. Interestingly, we found that awareness of this association was still low in participants with PD, although it was greater compared with overall awareness.

Our findings of a gap in awareness and safety practices related to pesticide use among people who use pesticides. Of the PPE types, gloves and masks were the most commonly used, each used by just over half of the respondents. The use of gloves while working with paraquat has been previously shown to modify or negate the risk of developing PD [6,11]. However it should be noted that non-use of certain items of PPE by respondents may not represent poor practice (i.e., the target usage for a given item is not necessarily 100%) because appropriate PPE will depend on the route of entry (e.g., skin or inhalation) for a given substance, whether it is used concentrated or diluted, and the type of formulation and application process. Disposal of PPE and user hygiene during and after use will also affect exposure levels. However, in this survey, 29% of respondents in contact with pesticides did not use any PPE at all. Several educational interventions have been shown to improve awareness of the effects of pesticides on health, and to subsequently improve safety behaviour and exposure mitigation [12]. The World Health Organization (WHO) recognises PPE as a method of exposure mitigation, along with other safety behaviours, and recommends government educational programmes for safe use of pesticides [20].

Our study participants identified herbicides as the most frequently encountered pesticide, followed by insecticides and fungicides. Both herbicides and insecticides have been linked to increased risk of PD [4,7,21,22]. Two specific classes of insecticides, organochlorines and organophosphorus compounds, have been significantly associated with the development of PD [19,23,24]. This finding is echoed in large-scale reviews of this research field [25–27]. The mechanism of action is more clearly understood for insecticides, which are purposefully designed to be acutely neurotoxic; they can achieve their toxicity in many specific ways that can affect the basic functions of neurons, including neurotransmission, mitochondrial function, and ion channel functioning [7,8,28,29]. Other pesticides, such as herbicides and fungicides, do not purposefully target nerve cells, but nevertheless have been found to affect cellular mechanisms that play roles in neurodegeneration, such as how cells release energy, process toxic waste products, or respond to inflammation [28,30,31].

Pesticide usage is documented by the Pesticide Registration and Control Division (PRCD) of the Department of Agriculture, Forestry and Marine through annual surveys across different sectors. For example, a 2020 survey of 464 grassland holdings found that 66 different substances were used, while a 2014 survey of 26 soft fruit holdings found that 41 active substances were used [32,33]. Safety regulations to protect workers, the environment, and the public from the potential hazards of pesticide use include Pesticide Regulations (S.I. No. 155/2012) [34], which implements the European Union's Sustainable Use of Pesticides Directive (2009/128/EC) [35] in Ireland. Irish farmers are obliged to be registered and to record the use of pesticides, and to ensure that their pesticide use practices are in line with the Irish National Action Plan for the Sustainable Use of Pesticides Directive [36], which aims to promote the responsible use of pesticides to reduce the impacts of these chemicals on human health and on the environment. A 2013 study reported that MCPA (herbicide), Glyphosate (herbicide), Chlorothalonil (fungicide), Mecoprop-P (herbicide), Chlormequat (plant growth regulator), and Mancozeb (fungicide) were the most commonly used pesticides in Ireland [37]. In another study, Glyphosate (herbicide) and Chlormequat-

chloride (plant growth regulator) were reported to be the most widely used, and were deemed to be the top pesticides of hazardous concern in Ireland [38]. Several agencies, such as the Health and Safety Authority, PRCD, and the Irish College of Agriculture and Food Science (Teagasc), provide materials, specific training, and certification pertaining to the safe use of pesticides. Many of these programmes focus on environmental and watercourse protection rather than personal safety. Nevertheless, there remains potential to further improve personal protection behaviours and to promote alternatives such as using biological controls, switching from organophosphates to permethrins, and introducing more sustainable agricultural practices alongside the reduction in pesticide use.

The purpose of this study was to document awareness of the link between pesticides and PD and to gather information on PPE use among farming communities. There are some limitations to consider. This study was not designed to be representative of all individuals occupationally exposed to pesticides. Because the respondents recruited were a convenience sample, they are unlikely to be representative of all in the farming community in Ireland. We are unable to validate or verify the accuracy of respondents' self-reported brands/classes of the pesticides that they use (respondents were relying on recall as they were not at their home or workplace at the time of completing the survey). In addition, we could not capture whether and how pesticide choice may have changed over time. Without in-depth interviews or substantially longer questionnaires, which could limit response rates, we could not ascertain the various types of pesticide application processes experienced by each individual; therefore, it may be the case that some non-use of certain PPE was indeed appropriate, rather than evidence of poor safety practice. Despite these limitations, we know of no reason why non-participants would be more likely to be aware of a link between pesticides and PD or more likely to use PPE. We also acknowledge that the initial question on awareness of links between pesticides and Parkinson's disease may have had a certain bias. However, this question did not influence the respondents' answers, since 75% of pesticides users in this sample were unaware of a link with PD, and 29% of users did not use any PPE. These are reasonable indicators that among the almost 300,000 farm workers in Ireland, many could adopt measures to further mitigate their risk.

5. Conclusions

This study highlights the urgent need for increased awareness and continuing education regarding pesticide use and its potential health risks, including PD, among farming communities in Ireland. In a highly agricultural economy which relies on pesticide use to optimise yield and profit, occupational exposure to pesticides is common. The dose of exposure is modifiable by the proper use of PPE. It is important to introduce comprehensive training programs that promote safer practices and to strengthen regulatory frameworks in order to mitigate these risks and improve the well-being of farmers and all people living in rural communities.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/safety11020049/s1>. Survey Questions.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in this study.

Data Availability Statement: The datasets generated during and/or analysed during the current study are available in the Zenodo repository, DOI: 10.5281/zenodo.14205980.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

DA	Dopamine
PD	Parkinson's disease
PPE	Personal Protective Equipment

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